

19

Equipartition of energy

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Before introducing the partition function in Chapter 20, which will allow us to calculate many different properties of thermodynamic systems on the basis of their microscopic energy levels (which can be deduced using quantum mechanics), we devote this chapter to the equipartition theorem. This theorem provides a simple, classical theory of thermal systems. It gives remarkably good answers, but only at high temperature, where the details of quantized energy levels can be safely ignored. We will motivate and prove this theorem in the following section, and then apply it to various physical situations in Section 19.2, demonstrating that it provides a rapid and straightforward method for deriving heat capacities. Finally, in Section 19.3, we will critically examine the assumptions which we have made in the derivation of the equipartition theorem.

19.1 Equipartition theorem

¹We will show later in Section 19.3 that this quadratic dependence is very common; most potential wells are approximately quadratic near the bottom of the well.

Very often in physics one is faced with an energy dependence which is quadratic in some variable.¹ An example would be the kinetic energy E_{KE} of a particle with mass m and velocity v , which is given by

$$E_{KE} = \frac{1}{2}mv^2. \quad (19.1)$$

Another example would be the potential energy E_{PE} of a mass suspended at one end of a spring with spring constant k and displaced by a distance x from its equilibrium point (see Figure 19.1). This is given by

$$E_{PE} = \frac{1}{2}kx^2. \quad (19.2)$$

In fact, the total energy E of a moving mass on the end of a spring is given by the sum of these two terms, so that

$$E = E_{KE} + E_{PE} = \frac{1}{2}mv^2 + \frac{1}{2}kx^2, \quad (19.3)$$

and, as the mass undergoes simple harmonic motion, energy is exchanged between E_{KE} and E_{PE} , while the total energy remains fixed.

Let us suppose that a system whose energy has a quadratic dependence on some variable is allowed to interact with a heat bath. It is then able to borrow energy occasionally from its environment, or even give it back into the environment. What mean thermal energy would it have? The

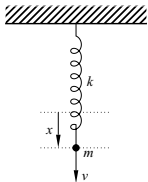


Fig. 19.1 A mass m suspended on a spring with spring constant k . The mass is displaced by a distance x from its equilibrium or 'rest' position.

thermal energy would be stored as kinetic or potential energy, so if a mass on a spring is allowed to come into thermal equilibrium with its environment, one could in principle take a very big magnifying glass and see the mass on a spring jiggling *around all by itself* owing to such thermal vibrations. How big would such vibrations be? The calculation is quite straightforward.

Let the energy E of a particular system be given by

$$E = \alpha x^2, \quad (19.4)$$

where α is some positive constant and x is some variable (see Fig. 19.2). Let us also assume that x could in principle take any value with equal probability. The probability $P(x)$ of the system having a particular energy αx^2 is proportional to the Boltzmann factor $e^{-\beta \alpha x^2}$ (see eqn 4.13), so that after normalizing, we have

$$P(x) = \frac{e^{-\beta \alpha x^2}}{\int_{-\infty}^{\infty} e^{-\beta \alpha x^2} dx}, \quad (19.5)$$

and the mean energy is

$$\begin{aligned} \langle E \rangle &= \int_{-\infty}^{\infty} E P(x) dx \\ &= \frac{\int_{-\infty}^{\infty} \alpha x^2 e^{-\beta \alpha x^2} dx}{\int_{-\infty}^{\infty} e^{-\beta \alpha x^2} dx} \\ &= \frac{1}{2\beta} \\ &= \frac{1}{2} k_B T. \end{aligned} \quad (19.6)$$

This is a really remarkable result. It is independent of the constant α and gives a mean energy which is proportional to temperature. The theorem can be extended straightforwardly to the energy being the sum of n quadratic terms, as shown in the following example.

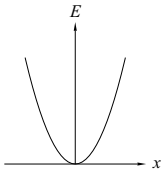


Fig. 19.2 The energy E of a system is $E = \alpha x^2$.

Example 19.1

Assume that the energy E of a system can be given by the sum of n independent quadratic terms, so that

$$E = \sum_{i=1}^n \alpha_i x_i^2, \quad (19.7)$$

where α_i are constants and x_i are some variables. Assume also that each x_i could in principle take any value with equal probability. Calculate the mean energy.

Solution:

The mean energy $\langle E \rangle$ is given by

$$\langle E \rangle = \int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} E P(x_1, x_2, \dots, x_n) dx_1 dx_2 \cdots dx_n. \quad (19.8)$$

This now now looks quite complicated when we substitute in the probability as follows

$$\langle E \rangle = \frac{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \left(\sum_{i=1}^n \alpha_i x_i^2 \right) \exp \left(-\beta \sum_{j=1}^n \alpha_j x_j^2 \right) dx_1 dx_2 \cdots dx_n}{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \exp \left(-\beta \sum_{j=1}^n \alpha_j x_j^2 \right) dx_1 dx_2 \cdots dx_n}, \quad (19.9)$$

where i and j have been used to distinguish different sums. This expression can be simplified by recognizing that it is the sum of n similar terms (write out the sums to convince yourself):

$$\langle E \rangle = \sum_{i=1}^n \frac{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \alpha_i x_i^2 \exp \left(-\beta \sum_{j=1}^n \alpha_j x_j^2 \right) dx_1 dx_2 \cdots dx_n}{\int_{-\infty}^{\infty} \cdots \int_{-\infty}^{\infty} \exp \left(-\beta \sum_{j=1}^n \alpha_j x_j^2 \right) dx_1 dx_2 \cdots dx_n}, \quad (19.10)$$

and then all but one integral cancels between the numerator and denominator of each term, so that

$$\langle E \rangle = \sum_{i=1}^n \frac{\int_{-\infty}^{\infty} \alpha_i x_i^2 \exp \left(-\beta \alpha_i x_i^2 \right) dx_i}{\int_{-\infty}^{\infty} \exp \left(-\beta \alpha_i x_i^2 \right) dx_i}. \quad (19.11)$$

Now each term in this sum is the same as the one treated above in eqn 19.6. Hence

$$\begin{aligned} \langle E \rangle &= \sum_{i=1}^n \alpha_i \langle x_i^2 \rangle = \sum_{i=1}^n \frac{1}{2} k_B T \\ &= \frac{n}{2} k_B T. \end{aligned} \quad (19.12)$$

Each quadratic energy dependence of the system is called a **mode** of the system (or sometimes a **degree of freedom** of the system). The spring, our example at the beginning of this chapter, has two such modes. The result of the example above shows that each mode of the system contributes an amount of energy equal to $\frac{1}{2} k_B T$ to the total mean energy of the system. This result is the basis of the **equipartition theorem**, which we state as follows:

Equipartition theorem:

If the energy of a classical system is the sum of n quadratic modes, and that system is in contact with a heat reservoir at temperature T , the mean energy of the system is given by $n \times \frac{1}{2} k_B T$.

The equipartition theorem expresses the fact that energy is ‘equally partitioned’ between all the separate modes of the system, each mode having a mean energy of precisely $\frac{1}{2} k_B T$.

Example 19.2

We return to our example of a mass on a spring, whose energy is given by the sum of two quadratic energy modes (see eqn 19.3). The equipartition theorem then implies that the mean energy is given by

$$2 \times \frac{1}{2} k_B T = k_B T. \quad (19.13)$$

How big is this energy? At room temperature, $k_B T \approx 4 \times 10^{-21}$ J $\approx$ 0.025 eV which is a tiny energy. This energy isn't going to set a 10 kg mass on a stiff spring vibrating very much! However, the extraordinary thing about the equipartition theorem is that the result holds *independently of the size of the system*, so that $k_B T = 0.025$ eV is also the mean energy of an atom on the end of a chemical bond (which can be modelled as a spring) at room temperature. For an atom, $k_B T = 0.025$ eV goes a very long way and this explains why atoms in molecules jiggle around a lot at room temperature. We will explore this in more detail below.

19.2 Applications

We now consider four applications of the equipartition theorem.

19.2.1 Translational motion in a monatomic gas

The energy of each atom in a monatomic gas is given by

$$E = \frac{1}{2} m v_x^2 + \frac{1}{2} m v_y^2 + \frac{1}{2} m v_z^2, \quad (19.14)$$

where $\mathbf{v} = (v_x, v_y, v_z)$ is the velocity of the atom (see Fig. 19.3). This energy is the sum of three independent quadratic modes, and thus the equipartition theorem gives the mean energy as

$$\langle E \rangle = 3 \times \frac{1}{2} k_B T = \frac{3}{2} k_B T. \quad (19.15)$$

This is in agreement with our earlier derivation of the mean kinetic energy of a gas (see eqn 5.17).

19.2.2 Rotational motion in a diatomic gas

In a diatomic gas, there is an additional possible energy source to consider, namely that of rotational kinetic energy. This adds two terms to the energy

$$\frac{L_1^2}{2I_1} + \frac{L_2^2}{2I_2}, \quad (19.16)$$

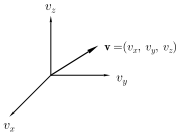


Fig. 19.3 The velocity of a molecule in a gas.

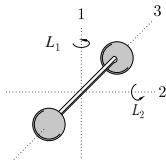


Fig. 19.4 Rotational motion in a diatomic gas.

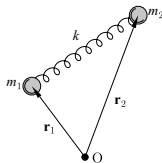


Fig. 19.5 A diatomic molecule can be modelled as two masses connected by a spring.

²See Appendix G.

where L_1 and L_2 are the angular momenta along the two principal directions shown in Fig. 19.4 and I_1 and I_2 are the corresponding moments of inertia. We do not need to worry about the direction along the diatomic molecule's bond, the axis labelled '3' in Fig. 19.4. (This is because the moment of inertia in this direction is very small (so that the corresponding rotational kinetic energy is very large), so rotational modes in this direction cannot be excited at ordinary temperature; such rotational modes are connected with the individual molecular electronic levels and we will therefore ignore them.)

The total energy is thus the sum of five terms, three due to translational kinetic energy and two due to rotational kinetic energy

$$E = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 + \frac{L_1^2}{2I_1} + \frac{L_2^2}{2I_2}, \quad (19.17)$$

and all of these energy modes are independent of one another. Using the equipartition theorem, we can immediately write down the mean energy as

$$\langle E \rangle = 5 \times \frac{1}{2}k_B T = \frac{5}{2}k_B T. \quad (19.18)$$

19.2.3 Vibrational motion in a diatomic gas

If we also include the vibrational motion of the bond linking the two atoms in our diatomic molecule, there are two additional modes to include. The intramolecular bond can be modelled as a spring (see Fig. 19.5), so that the two extra energy terms are the kinetic energy due to *relative* motion of the two atoms and the potential energy in the bond (let us suppose it has spring constant k). Writing the positions of the two atoms as $\mathbf{r}_1$ and $\mathbf{r}_2$ with respect to some fixed origin, the energy of the atom can be written

$$E = \frac{1}{2}mv_x^2 + \frac{1}{2}mv_y^2 + \frac{1}{2}mv_z^2 + \frac{L_1^2}{2I_1} + \frac{L_2^2}{2I_2} + \frac{1}{2}\mu(\dot{\mathbf{r}}_1 - \dot{\mathbf{r}}_2)^2 + \frac{1}{2}k(\mathbf{r}_1 - \mathbf{r}_2)^2, \quad (19.19)$$

where $\mu = m_1 m_2 / (m_1 + m_2)$ is the reduced mass² of the system. The equipartition theorem just cares about the number of modes in the system, so the mean energy is simply

$$\langle E \rangle = 7 \times \frac{1}{2}k_B T = \frac{7}{2}k_B T. \quad (19.20)$$

The heat capacity of the systems described above can be obtained by differentiating the energy with respect to temperature. The mean energy is given by

$$\langle E \rangle = \frac{f}{2}k_B T, \quad (19.21)$$

where f is the number of degrees of freedom. This equation implies that

$$C_V \text{ per mole} = \frac{f}{2}R, \quad (19.22)$$

and using eqn 11.28 we have

$$C_p \text{ per mole} = \left(\frac{f}{2} + 1 \right) R, \quad (19.23)$$

from which we may derive

$$\gamma = \frac{C_p}{C_V} = \frac{(\frac{f}{2} + 1)R}{\frac{f}{2}R} = 1 + \frac{2}{f}. \quad (19.24)$$

We can summarize our results for the heat capacity of gases, per atom/molecule, as follows:

Gas	Modes	f	$\langle E \rangle$	γ
Monatomic	translational only	3	$\frac{3}{2}k_B$	$\frac{5}{3}$
Diatomic	translational and rotational	5	$\frac{5}{2}k_B$	$\frac{7}{5}$
Diatomic	translational, rotational and vibrational	7	$\frac{7}{2}k_B$	$\frac{9}{7}$

19.2.4 The heat capacity of a solid

In a solid, the atoms are held rigidly in the lattice and there is no possibility of translational motion. However, the atoms can vibrate about their mean positions. Consider a cubic solid in which each atom is connected by springs (chemical bonds) to six neighbours (one above, one below, one in front, one behind, one to the right, one to the left). Since each spring joins two atoms, then if there are N atoms in the solid, there are $3N$ springs (neglecting the surface of the solid, a reasonable approximation if N is large). Each spring has two quadratic modes of energy (one kinetic, one potential) and hence a mean thermal energy equal to $2 \times \frac{1}{2}k_B T = k_B T$. Hence the mean energy of the solid is

$$\langle E \rangle = 3Nk_B T, \quad (19.25)$$

and the heat capacity is $\partial \langle E \rangle / \partial T = 3Nk_B$. Because $R = N_A k_B$, the molar heat capacity of a solid is then expected to be $3N_A k_B = 3R$.

19.3 Assumptions made

The equipartition theorem seems to be an extremely powerful tool for evaluating thermal energies of systems. However, it does have some limitations, and to discover what these are, it is worth thinking about the assumptions we have made in deriving it.

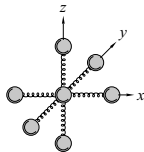


Fig. 19.6 In a cubic solid, each atom is connected by chemical bonds, modelled as springs, to six nearest neighbours, two along each of the three Cartesian axes. Each spring is shared between two atoms.

- We have assumed that the parameter for which we have taken the energy to be quadratic can take any possible value. In the derivation, the variables x_i could be integrated continuously from $-\infty$ to ∞ . However, quantum mechanics insists that certain quantities can only take particular ‘quantized’ values. For example, the problem of a mass on a spring is shown by quantum mechanics to have an energy spectrum which is quantized into levels given by $(n + \frac{1}{2})\hbar\omega$. When the thermal energy $k_B T$ is of the same order, or lower than, $\hbar\omega$, the approximation made by ignoring the quantized nature of this energy spectrum is going to be a very bad one. However, when $k_B T \gg \hbar\omega$, the quantized nature of the energy spectrum is going to be largely irrelevant, in much the same way that you don’t notice that the different shades of grey in a newspaper photograph are actually made up of lots of little dots if you don’t look closely. Thus we come to an important conclusion:

The equipartition theorem is generally valid only at *high temperature*, so that the thermal energy is larger than the energy gap between quantized energy levels. Results based on the equipartition theorem should emerge as the high-temperature limit of more detailed theories.

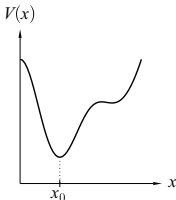


Fig. 19.7 $V(x)$ is a function which is more complicated than a quadratic but which has a minimum at $x = x_0$.

³Using a Taylor expansion; see Appendix B.

⁴The argument that the bottom of almost all potential wells tends to be approximately quadratic could fail if $(\partial^2 V / \partial x^2)_{x_0}$ turned out to be zero. This would happen if, for example, $V(x) = \alpha(x - x_0)^4$.

- Everywhere we have assumed that modes are quadratic. Is that always valid? To give a concrete example, imagine that an atom moves with coordinate x in a potential well given by $V(x)$, which is a function which might be more complicated than a quadratic (see for example Fig. 19.7). At absolute zero, the atom finds a potential minimum at say x_0 (so that, for the usual reasons, $\partial V / \partial x = 0$ and $\partial^2 V / \partial x^2 > 0$ at $x = x_0$). At temperature $T > 0$, the atom can explore regions away from x_0 by borrowing energy of order $k_B T$ from its environment. Near x_0 , the potential $V(x)$ can be expanded³ as

$$V(x) = V(x_0) + \left(\frac{\partial V}{\partial x} \right)_{x_0} (x - x_0) + \frac{1}{2} \left(\frac{\partial^2 V}{\partial x^2} \right)_{x_0} (x - x_0)^2 + \dots, \quad (19.26)$$

so that using $(\partial V / \partial x)_{x_0} = 0$, we find that the potential energy is

$$V(x) = \text{constant} + \frac{1}{2} \left(\frac{\partial^2 V}{\partial x^2} \right)_{x_0} (x - x_0)^2 + \dots, \quad (19.27)$$

which is a quadratic again. This demonstrates that the bottom of almost all potential wells tends to be approximately quadratic (this is known as the **harmonic approximation**).⁴ If the temperature gets too high, the system will be able to access positions far away from x_0 and the approximation of ignoring the higher-order (cubic, quartic, etc.) terms (known as the **anharmonic terms**) in the Taylor expansion may become important.

However, we have just said that the equipartition theorem is only valid at high temperature. Thus we see that the temperature must be high enough that we can safely ignore the quantum nature of the energy spectrum, but not so high that we invalidate the approximation of treating the relevant potential wells as perfectly quadratic. Fortunately there is plenty of room between these two extremes.

19.4 Brownian motion

We close this chapter with one example in which the effect of the equipartition of energy is encountered.

Example 19.3

Brownian motion:

In 1827, Robert Brown used a microscope to observe pollen grains jiggling about in water. He was not the first to make such an observation (any small particles suspended in a fluid will do the same, and are very apparent when looking down a microscope), but this effect has come to be known as **Brownian motion**.

The motion is very irregular, consisting of translations and rotations, with grains moving independently, even when moving close to each other. The motion is found to be more active the smaller the particles. The motion is also found to be more active the less viscous the fluid. Brown was able to discount a 'vital' explanation of the effect, i.e. that the pollen grains were somehow 'alive', but he was not able to give a correct explanation. Something resembling a modern theory of Brownian motion was proposed by Wiener in 1863, though the major breakthrough was made by Einstein in 1905.

We will postpone a full discussion of Brownian motion until Chapter 33, but using the equipartition theorem, the origin of the effect can be understood in outline. Each pollen grain (of mass m) is free to move translationally and so has mean kinetic energy $\frac{1}{2}m\langle v^2 \rangle = \frac{3}{2}k_B T$. This energy is very small, as we have seen, but leads to a measurable amplitude of vibration for a small pollen grain. The amplitude of vibration is greater for smaller pollen grains because a mean kinetic energy of $\frac{3}{2}k_B T$ gives more mean square velocity ($\langle v^2 \rangle$) to less massive grains. The thermally excited vibrations are resisted by viscous damping, so the motion is expected to be more pronounced in less viscous fluids.

Chapter summary

- The equipartition theorem states that if the energy of a system is the sum of n quadratic modes, and that the system is in contact with a heat reservoir of temperature T , the mean energy of the system is given by $n \times \frac{1}{2}k_B T$.
- The equipartition theorem is a high-temperature result and gives incorrect predictions at low temperature, where the discrete nature of the energy spectrum cannot be ignored.

Exercises

- (19.1) What is the mean kinetic energy in eV at room temperature of a gaseous (a) He atom, (b) Xe atom, (c) Ar atom and (d) Kr atom. [Hint: do you have to do four separate calculations?]
- (19.2) Comment on the following values of molar heat capacity in $\text{J K}^{-1} \text{mol}^{-1}$, all measured at constant pressure at 298 K.

Al	24.35	Pb	26.44
Ar	20.79	Ne	20.79
Au	25.42	N ₂	29.13
Cu	24.44	O ₂	29.36
He	20.79	Ag	25.53
H ₂	28.82	Xe	20.79
Fe	25.10	Zn	25.40

[Hint: express them in terms of R ; which of the substances is a solid and which is gaseous?]

- (19.3) A particle at position r is in a potential well $V(r)$ given by

$$V(r) = \frac{A}{r^n} - \frac{B}{r}, \quad (19.28)$$

where A and B are positive constants and $n > 2$. Show that the bottom of the well is approximately quadratic in r . Hence find the particle's mean thermal energy at temperature T above the bottom of the well assuming the validity of the equipartition theorem in this situation.

- (19.4) In example 19.1, show that

$$\langle x_i^2 \rangle = \frac{k_B T}{2\alpha_i}. \quad (19.29)$$

- (19.5) If the energy E of a system is not quadratic, but behaves like $E = \alpha|x|$ where $\alpha > 0$, show that $\langle E \rangle = k_B T$.
- (19.6) If the energy E of a system behaves like $E = \alpha|x|^n$, where $n = 1, 2, 3, \dots$ and $\alpha > 0$, show that $\langle E \rangle = \xi k_B T$, where ξ is a numerical constant.
- (19.7) A simple pendulum with length ℓ makes an angle θ with the vertical, where $\theta \ll 1$. Show that it oscillates with a period given by $2\pi\sqrt{\ell/g}$. The pendulum is now placed at rest and allowed to come into equilibrium with its surroundings at temperature T . Derive an expression for $\langle \theta^2 \rangle$.